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Robust Personalized Federated Learning with Sparse Penalization

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ABSTRACT

Federated learning (FL) is an emerging topic due to its advantage in collaborative learning with distributed data. Due to the heterogeneity in the local data-generating mechanism, it is important to consider personalization when developing federated learning methods. In this work, we propose a personalized federated learning (PFL) method to address the robust regression problem. Specifically, we aim to learn the regression weight by solving a Huber loss with the sparse fused penalty. Additionally, we designed our personalized federated learning for robust and sparse regression (PerFL-RSR) algorithm to solve the estimation problem in the federated system efficiently. Theoretically, we show that the proposed PerFL-RSR reaches a convergence rate of $\mathcal{O}(1/T)$, and the proposed estimator is statistically consistent. Thorough experiments and real data analysis are conducted to corroborate the theoretical results of our proposed personalized federated learning method. Supplementary materials for this article are available online including a standardized description of the materials available for reproducing the work.

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Federated learning;
Heterogeneous data;
Personalization; Robust
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1. Introduction

Federated learning (FL) has received increasing research attention recently due to its success in distributed machine learning tasks (McMahan et al. 2017; Wu et al. 2022). In an FL system, a large number of clients, which could be mobile phones, wireless sensors, medical facilities, autonomous vehicles, etc., will collaboratively train a machine learning model coordinated by a central server. Rather than transferring all the data to the central server as in the traditional distributed learning system, FL allows the clients to keep their local data and thus gain a basic level of data privacy. Since the inception, FL has exhibited its advantages in model learning accuracy and computation efficiency due to numerous clients and massive data sources. However, due to the heterogeneous data-generating mechanism among the clients, there emerge two new challenges that might cause performance deterioration on FL: statistical heterogeneity and system heterogeneity.

It has been observed that heterogeneous data (i.e., non-iid data) is prevalent in real-world applications. For example, medical images from different hospitals may vary due to different collecting devices and imaging protocols. Data heterogeneity often causes performance degradation in FL (McMahan et al. 2017; Khaled, Mishchenko, and Richtárik 2020). Many recent studies proposed various technologies to tackle the data heterogeneity issue in FL, such as feature normalization (Huang and Belongie 2017; Choi et al. 2021), model weights regularization (Karimireddy et al. 2020), and improving aggregation (Reddi et al. 2020; Wang et al. 2020). However, the above methods focus on training a global shared model, which means that they

assume a common global model for all the clients. The global model-sharing assumption might not be appropriate, especially when data distributions differ significantly among clients. The study (Yu, Bagdasaryan, and Shmatikov 2020) shows that model performance would regress when switching from locally learned models to a global shared model. In other words, the traditional FL with the global model sharing assumption would sacrifice the generalizability of the model. In real-world machine learning tasks, personalization is often a topic of great interest. For instance, precision medicine pursues a customized model for each patient.

To balance generalization and personalization, personalized federated learning (PFL) is proposed, which focuses on learning personalized models for each client. Figure 1 shows the architectures and communications for FL and PFL, respectively. Rather than aggregation and generation of one global model in FL, in PFL, the server needs to learn the relationship or similarity between local models and then generate personalized models for each client. Many recent studies proposed various methods for PFL, such as fine-tuning (Fallah, Mokhtari, and Ozdaglar 2020), multi-task learning (MTL) (Smith et al. 2017; Huang et al. 2021), clustered federated learning (Ghosh et al. 2020; Sattler, Müller, and Samek 2020), parameter decoupling (Arivazhagan et al. 2019) and knowledge distillation (Li and Wang 2019) etc. In this work, we will combine the idea of MTL and clustered FL. MTL achieves personalization by studying the similarity between local client models, and clustered FL achieves personalization through inherent partitions of all local client models. Rather than in most MTL methods, the similarity

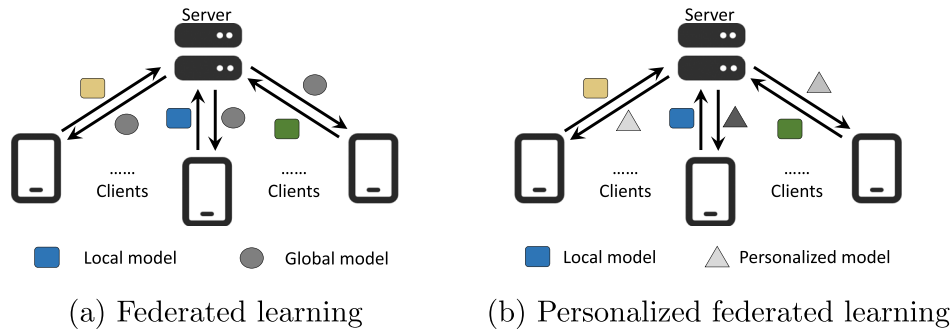


Figure 1. Architectures and communications for federated learning and personalized federated learning, a. Federated learning, b. Personalized federated learning.

between local models and a defined global model is studied, and we aim to find out the relationships of each weight among clients, and thus the clients could be partitioned by each weight. The details of the model weights structure are introduced in Section 3.

System heterogeneity becomes a considerable bottleneck in federated learning (McMahan et al. 2017; Smith et al. 2017; Ghosh et al. 2020; Fallah, Mokhtari, and Ozdaglar 2020; Wang, Marella, and Anderson 2022), especially when there are a large number of local clients in the network, such as learning over mobile phones, wearable devices, and autonomous vehicles. Because of the different conditions of the clients, for example the network connection and power status of the devices, it may be impractical to involve all the clients in each communication round. Hence, we consider designing a federated learning algorithm that allows a low participation rate per model iteration round.

In this work, we will focus on personalized federated learning for robust regression problems. Our study is motivated by the fact that many real-world datasets are contaminated with heavy-tailed noises and abnormal values, especially for the federated learning system with massive data sources. Furthermore, we consider that the regression problem would be high-dimensional, which is also common in massive data sources. The high dimensionality issue necessitates the model sparsity recovery. Therefore, in this work, our objective is to develop a personalized federated learning method for robust and sparse regression. Our main contributions and their significance are summarized in the following:

- We balance the tradeoff of personalization and generalization for the robust sparse regression problem under the federated learning paradigm. We propose a novel learning loss (1), which consists of the Huber loss for robustness, the client-wise fusion regularizer for personalization, and the sparse regularizer for sparsity recovery. After solving the proposed loss, each client can obtain a personalized model that integrates useful information from similar data sources.
- We develop an alternating direction method of multipliers (ADMM) based algorithm in the federated server-client system to solve the proposed loss (1). We name the proposed algorithm as personalized federated learning for robust and sparse regression (PerFL-RSR). PerFL-RSR addresses system heterogeneity and communication efficiency through random sampling clients for each update on the server.

By leveraging the linear approximation of the Huber loss, PerFL-RSR improves computation efficiency on the clients, and thus reduces communication rounds.

- We establish the convergence theory for the proposed PerFL-RSR algorithm. The augmented Lagrangian for the proposed problem does not satisfy the assumptions for the state-of-the-art convergence theories of linearized ADMM, such as Li and Pong (2015), Huang, Chen, and Huang (2019), and Lu et al. (2021). We extend the convergence of linearized ADMM and show that our PerFL-RSR converges with the rate of $\mathcal{O}(1/T)$.
- Furthermore, we establish the statistical consistency properties for our proposed estimator obtained from solving (1). Under some common assumptions, we show the statistical accuracy of the oracle estimator and that the oracle estimator is a local minimizer of the proposed concave objective function. Furthermore, we derive the upper error bound and support recovery property of any local minimum of the proposed concave objective function.

The rest of the article is organized as follows. In Section 2, we go over the related works on robust linear regression and personalized federated learning. We formulate the robust regression problem under the PFL context in Section 3 and propose our PerFL-RSR algorithm in Section 4. We establish the theoretical proprieties of our estimator in Section 5. Simulation and real data analysis are in Section 6. We conclude the article in Section 7. All the technical details are relegated to supplementary materials.

2. Related Works

2.1. Personalized Federated Learning

There are various approaches to personalized federated learning. Recent PFL methods such as Fallah, Mokhtari, and Ozdaglar (2020) and Wang, Marella, and Anderson (2022) and sparse PFL methods such as Dai et al. (2022), and Liu et al. (2023) studied personalized local models that are close to the global model. However, our aim is to obtain efficiency by investigating the similarity between personalized local models. Here, we review two related approaches: federated multi-task learning and clustered federated learning.

Federated multi-task learning is an effective solution for training personalized models jointly for all clients by treating each client as a task, which improves estimation accuracy by

sharing information across clients. The MOCHA algorithm (Smith et al. 2017) has been proposed to solve MTL in FL settings. FedAMP (Huang et al. 2021) proposed a message-passing mechanism to solve loss with pair-wise regularization terms among all clients. However, MOCHA and FedAMP are designed for convex objectives and are not applicable to our proposed non-convex loss.

Clustering FL assumes that there are homogeneous groups of clients in terms of local data distributions. IFCA (Ghosh et al. 2020) has been proposed to learn K global models on the server, and then each client can choose a model with the smallest local loss. However, for IFCA, the server needs to communicate K times more information, and prior knowledge of the number of groups K is required. In Sattler, Müller, and Samek (2020), authors proposed an algorithm with a post-processing step of clustering. They use recursive bi-partitioning clustering, which increases computation and communication costs. For regression models, several works (Ma and Huang 2017; Ma et al. 2019; Yang, Yan, and Huang 2019; Austin, Pan, and Shen 2020) studied user-specific regression models with square loss. However, they did not consider model robustness, and their algorithms are not applicable in the FL setting.

Due to the limitations above of the existing works, there is no trivial extension to the proposed sparse fused Huber loss (1) in the federated learning framework.

2.2. Robust Regression

The regression model has been the most widely adopted analysis technique for data fitting and prediction in the past few decades. The standard procedure is to estimate the regression weights by minimizing the ordinary least square loss, that is, $\arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2$. However, the estimator performs poorly in the presence of heavy-tailed data and data with outliers. Therefore, the robust regression method is of increasing research interest.

The median-on-means method is one of the common ways to handle heavy-tailed data and enhance the robustness of regression (Lecué and Lerasle 2019; Lugosi and Mendelson 2019; Lecué and Lerasle 2020). However, the median-on-means method requires random data splitting, which is not suitable in the federated learning framework where the data is preserved by each data provider. Moreover, when it comes to sparse regression problems, the support recovery performance of the median on means method remains unclear.

Another robust regression method is the quantile regression loss (Zou and Yuan 2008; Fan, Fan, and Barut 2014; Chen et al. 2020). Note that when the quantile level is taken as 1/2, the quantile loss becomes the absolute deviation loss. Although various algorithms have been proposed to solve it, such as greedy coordinate descent algorithm (Peng and Wang 2015), subgradient descent algorithm (Wang and Li 2017) or ADMM algorithm (Gu et al. 2018), most of them either lack theoretical convergence guarantee or suffer from a slow convergence rate due to the non-smoothness nature of the quantile loss.

In the literature, Huber loss Huber (1964) has been used to improve learning robustness thanks to its smoothness property and advantage in optimization. Huber loss combines the square

loss and the absolute deviation loss through a blending parameter σ . In Huber (1973) and Yohai and Maronna (1979), the authors have studied the properties of the estimator from Huber loss. In Sun, Zhou, and Fan (2020), the authors showed the non-asymptotic robustness of adaptive Huber regression for errors with exponential-type concentration bounds when only low-order moments exist. Based on the smoothness of Huber loss, Sun, Zhou, and Fan (2020) suggested using the local adaptive majorize-minimization algorithm (Fan et al. 2018) for optimization. In this work, we achieve federated learning robustness by adopting Huber loss and finding a good balance between personalization and generalization.

3. Problem Setup

In this section, we introduce the personalized federated problem for sparse and robust regression. For notation simplicity, let $[n]$ denote the set $\{1, \dots, n\}$. Let the regular letter denote scalar, the bold lower letter denote vector, and the bold upper letter denote matrix.

We consider a federation of M clients. Each client owns a local dataset $\mathcal{D}_m = \{\mathbf{x}_{mi}, y_{mi}\}_{i=1}^{n_m}$, for $m = 1, \dots, M$, where y is the response variable and $\mathbf{x} = (x_1, \dots, x_p)^\top \in \mathbb{R}^p$ is a p -dimensional covariate vector. Considering the data heterogeneity, we assume client-specific linear models: $y_{mi} = \mathbf{x}_{mi}^\top \beta_m^* + \epsilon_{mi}$, $\forall i \in [n_m], m \in [M]$, where $\beta_m^* = (\beta_{m1}^*, \dots, \beta_{mp}^*)^\top \in \mathbb{R}^p$ is the true regression weight for the m th client and might be different across the clients. The independent random error ϵ_{mi} has $\mathbb{E}(\epsilon_{mi} | \mathbf{x}_{mi}) = 0$ and $\mathbb{E}(|\epsilon_{mi}|^{1+\delta}) < \infty$ for some $\delta > 0$. Note that, because of the data heterogeneity, the covariance structure $\mathbb{E}_{\mathbf{x}_{mi} \in \mathcal{D}_m}(\mathbf{x}_{mi} \mathbf{x}_{mi}^\top)$ and the distributions of the random error ϵ_{mi} across the clients could be different.

First, in this work, we address model personalization and hence allow client-specific model weights β_m , $m \in [M]$, which could be estimated from $\arg \min_{\beta_1, \dots, \beta_M \in \mathbb{R}^p} \frac{1}{M} \sum_{m=1}^M l_m(\beta_m)$.

Next, we aim to find the similarity of weights between clients to take advantage of training with joint data. In clustered FL and MTL, the similarity is measured between the whole vector of weights. Here, we consider a more flexible structure of weights such that each coordinate of the weights might form a cluster structure. Hence, the weights structure of the clustered FL is a special case of the proposed method such that all the coordinates form the same clusters. We adopt the regularization term on the scalar-wise and pair-wise differences on β_{mj} and $\beta_{m'j}$ for $j \in [p]$. Then, our PFL solution is obtained by solving $\arg \min_{\beta_1, \dots, \beta_M \in \mathbb{R}^p} \frac{1}{M} \sum_{m=1}^M l_m(\beta_m) + \sum_{j=1}^p \sum_{m \leq m'} p_\lambda(|\beta_{mj} - \beta_{m'j}|)$, where $p_\lambda(\cdot)$ is a regularizer to be specified later, and $\lambda > 0$ is the regularization parameter.

Last, to achieve the model robustness with respect to outliers and heavy-tailed data, we adopt the Huber loss (Huber 1964) $H_\sigma(x) = \begin{cases} x^2/2, & \text{if } |x| \leq \sigma \\ \sigma|x| - \sigma^2/2, & \text{if } |x| > \sigma, \end{cases}$ where σ controls the blending of quadratic and ℓ_1 losses. Compared with the least square loss, the Huber loss down-weights the outlier data and thus leads to robustness. For ease of presentation, we vectorize the M model weights as $\beta = \text{vec}[(\beta_1, \dots, \beta_M)] \in \mathbb{R}^{pM}$.

To summarize, we achieve PFL for sparse and robust regression by solving

$$\begin{aligned} \hat{\beta} = \arg \min_{\beta \in \mathbb{R}^{pM}} & \frac{1}{M} \sum_{m=1}^M \frac{1}{n_m} \sum_{i=1}^{n_m} H_{\sigma} (y_{mi} - \mathbf{x}_{mi}^{\top} \beta_m) \\ & + \sum_{j=1}^p \sum_{m \leq m'} p_{\lambda_1} (|\beta_{mj} - \beta_{m'j}|) + \sum_{j=1}^p \sum_{m=1}^M p_{\lambda_2} (|\beta_{mj}|), \end{aligned} \quad (1)$$

in which $p_{\lambda}(\cdot)$ is the regularization term with the regularization parameter $\lambda > 0$ and $\lambda = (\lambda_1, \lambda_2)$. The first regularization term is used to balance the personalization and similarity between β_m , and the second regularization term is for sparsity recovery.

Here, we consider two concave regularizers: SCAD (Fan and Li 2001) and MCP (Zhang 2010), which enjoy a better sparsity recovery performance than ℓ_1 regularizer and reduce the estimation bias compared to ℓ_1 regularizer. For SCAD, $p_{\lambda}^{\text{SCAD}}(x) =$

$$\begin{cases} \lambda|x|, & \text{if } |x| \leq \lambda, \\ \frac{\iota\lambda|x| - x^2/2 - \lambda^2/2}{\iota - 1}, & \text{if } \lambda < |x| \leq \lambda\iota, \text{ where } \iota > 2. \\ \frac{\lambda^2(\iota+1)}{2}, & \text{if } |x| > \lambda\iota, \end{cases}$$

$$p_{\lambda}^{\text{MCP}}(x) = \begin{cases} \lambda|x| - \frac{x^2}{2\iota}, & \text{if } |x| \leq \lambda\iota, \\ \frac{\lambda^2}{2}, & \text{if } |x| > \lambda\iota, \end{cases} \text{ where } \iota > 1.$$

The proposed estimator $\hat{\beta}$ obtained from (1) consists of M personalized models, which enjoy integrated data information from all clients. Besides, the personalized model for each client has sparsity recovery so the proposed method works for data with a large number of features and could identify significant features. Moreover, the proposed estimator $\hat{\beta}$ has a group structure for each coordinate, which enables the investigation of the similarity between models.

4. Proposed Method

In this section, we introduce the proposed algorithm for personalized federated learning for robust and sparse regression (PerFL-RSR).

Notation: To measure the model dissimilarity, we introduce a matrix $\Omega = \mathbf{E} \otimes \mathbf{I}_p$, where $\mathbf{E} = (\mathbf{e}_i - \mathbf{e}_j, i < j)^{\top} \in \mathbb{R}^{\frac{M(M-1)}{2} \times M}$ with the standard basis vectors $\{\mathbf{e}_i\} \subset \mathbb{R}^M$, $\mathbf{I}_p$ is the $p \times p$ identity matrix and $\otimes$ is the Kronecker product. For any vector $\mathbf{x} \in \mathbb{R}^p$, we use $\|\mathbf{x}\|_2 = (\mathbf{x}^{\top} \mathbf{x})^{1/2}$ to denote the ℓ_2 -norm and $\|\mathbf{x}\|_1 = \sum_{i=1}^p |x_i|$ to denote the ℓ_1 -norm. For any positive definite matrix $\mathbf{H}$, we denote $\|\mathbf{x}\|_{\mathbf{H}}^2 = \mathbf{x}^{\top} \mathbf{H} \mathbf{x}$. For any $p \times p$ matrix $\mathbf{M}$, we use $\varsigma_{\min}(\mathbf{M})$ to denote the smallest eigenvalue of $\mathbf{M}$ and $\varsigma_{\max}(\mathbf{M})$ to denote the largest eigenvalue of $\mathbf{M}$. Let $\mathbb{I}[\cdot]$ be the indicator function. Let $\nabla f(\omega)$ denote the gradient of a function f with respect to the model weight ω . We say $a_n \lesssim b_n$ if $a_n \leq cb_n$ for some constant c , $a_n \gtrsim b_n$ if $a_n \geq c'b_n$ for some constant c' , and $a_n \asymp b_n$ when $a_n \leq cb_n$ and $a_n \geq c'b_n$ both hold.

4.1. Problem Reformulation

Letting $\mathbf{A} = [\Omega^{\top}, \mathbf{I}_{p \times M}]^{\top}$ and $\delta = \mathbf{A}\beta$, problem (1) can be reformulated as a constraint optimization problem:

$$\min f_{\sigma}(\beta) + h_{\lambda}(\delta), \text{ s.t. } \mathbf{A}\beta = \delta,$$

where $f_{\sigma}(\beta) = M^{-1} \sum_{m=1}^M n_m^{-1} \sum_{i=1}^{n_m} H_{\sigma}(y_{mi} - \mathbf{x}_{mi}^{\top} \beta_m)$, $h_{\lambda}(\delta) = \sum_d p_{\lambda^*}(|\delta_d|)$ and λ^* is λ_2 if $d > pM(M-1)/2$ and is λ_1 otherwise.

Augmented Lagrangian: Typically, the above problem can be solved by adopting ADMM based algorithm on its augmented Lagrangian:

$$\mathcal{L}_{\rho}(\beta, \delta, \gamma) = f_{\sigma}(\beta) + h_{\lambda}(\delta) + \langle \gamma, \mathbf{A}\beta - \delta \rangle + \frac{\rho}{2} \|\mathbf{A}\beta - \delta\|_2^2, \quad (2)$$

where γ is the Lagrange multiplier and ρ is the penalty parameter. Specifically, at t th iteration, β , δ , and γ can be sequentially updated as

$$\beta^{(t+1)} = \arg \min_{\beta} \mathcal{L}_{\rho}(\beta, \delta^{(t)}, \gamma^{(t)}), \quad (3)$$

$$\delta^{(t+1)} = \arg \min_{\delta} \mathcal{L}_{\rho}(\beta^{(t+1)}, \delta, \gamma^{(t)}),$$

$$\gamma^{(t+1)} = \gamma^{(t)} + \rho(\mathbf{A}\beta^{(t+1)} - \delta^{(t+1)}), \quad (4)$$

The update of Lagrange multiplier γ is straightforward.

Update of δ : Parameter δ is updated in a deterministic way through the thresholding operators defined by SCAD and MCP as follows. Let the soft thresholding rule be $ST(x, \lambda) = \text{sgn}(x)(|x| - \lambda)\mathbb{I}[|x| - \lambda > 0]$. Let $T_{\lambda, \rho}$ denote the thresholding operator defined as $T_{\lambda, \rho}(y) = \arg \min_{x \in \mathbb{R}} \{p_{\lambda}(x) + \rho/2(x - y)^2\}$. Then, the thresholding operator for SCAD regularizer is

$$T_{\lambda, \rho}^{\text{SCAD}}(y) = \begin{cases} ST(y, \lambda/\rho), & \text{if } |y| \leq \lambda + \frac{\lambda}{\rho}, \\ \frac{(\iota\rho - \rho)ST(y, \frac{\lambda}{\iota\rho - \rho})}{\iota\rho - \rho - 1}, & \text{if } \lambda + \frac{\lambda}{\rho} < |y| \leq \lambda\iota, \\ y, & \text{if } |y| > \lambda\iota, \end{cases} \text{ and}$$

the thresholding operator for MCP regularizer is $T_{\lambda, \rho}^{\text{MCP}}(y) = \begin{cases} \frac{\iota\rho}{\iota\rho - 1} ST(y, \lambda/\rho), & \text{if } |y| \leq \lambda\iota, \\ y, & \text{if } |y| > \lambda\iota. \end{cases}$ Therefore, the d th element in $\delta^{(t+1)}$ is updated as

$$\delta_d^{(t+1)} = T_{\lambda^*, \rho} \left(\left[\mathbf{A}\beta^{(t+1)} + \rho^{-1} \gamma^{(t)} \right]_d \right), \quad (5)$$

where λ^* is λ_2 if $d > pM(M-1)/2$ and is λ_1 otherwise.

Update of β : The update of β is not a trivial problem. By taking advantage of using square loss, Ma et al. (2019) and Yang, Yan, and Huang (2019) solved step (3) explicitly. However, there is no closed-form solution for the Huber loss. FedADMM updated local models inexactly such that $\|\nabla_{\beta_m} \mathcal{L}\|^2 \leq \varepsilon_m$. However, the final convergence rate has an extra term introduced by the local solution accuracy and hence would increase the communication rounds. Thus, we consider the linearized method for the ADMM and define an approximation function of the form: $\tilde{\mathcal{L}}(\beta; \beta^{(t)}, \delta^{(t)}, \gamma^{(t)}) = f_{\sigma}(\beta^{(t)}) + \nabla f_{\sigma}(\beta^{(t)})^{\top} (\beta - \beta^{(t)}) + \frac{1}{2\tau} \|\beta - \beta^{(t)}\|_{\mathbf{H}}^2 + \langle \gamma, \mathbf{A}\beta - \delta \rangle + \frac{\rho}{2} \|\mathbf{A}\beta - \delta\|_2^2$, where $\tau > 0$, $\mathbf{H}$ is a positive definite matrix. By minimizing $\tilde{\mathcal{L}}(\beta; \beta^{(t)}, \delta^{(t)}, \gamma^{(t)})$ with respect to β , we obtain $\beta^{(t+1)} = (\tau^{-1} \mathbf{H} + \rho \mathbf{A}^{\top} \mathbf{A})^{-1} [\tau^{-1} \mathbf{H} \beta^{(t)} - \nabla f_{\sigma}(\beta^{(t)}) + \rho \mathbf{A}^{\top} \delta^{(t)} - \mathbf{A}^{\top} \gamma^{(t)}]$. Since $\tau^{-1} \mathbf{H} + \rho \mathbf{A}^{\top} \mathbf{A}$ is a $Mp \times Mp$ matrix, when the number of data units or the number of variables is large, calculating the inverse $(\tau^{-1} \mathbf{H} + \rho \mathbf{A}^{\top} \mathbf{A})^{-1}$ is expensive in computing and storage. To avoid the problem, we use $\mathbf{H} = r\mathbf{I} - \rho\tau \mathbf{A}^{\top} \mathbf{A}$, and $r \geq \rho\tau \cdot \varsigma_{\max}(\mathbf{A}^{\top} \mathbf{A}) + 1$. Then, we have

$$\beta^{(t+1)} = r^{-1} \mathbf{H} \beta^{(t)} - r^{-1} \tau [\nabla f_{\sigma}(\beta^{(t)}) - \rho \mathbf{A}^{\top} \delta^{(t)} + \mathbf{A}^{\top} \gamma^{(t)}]. \quad (6)$$

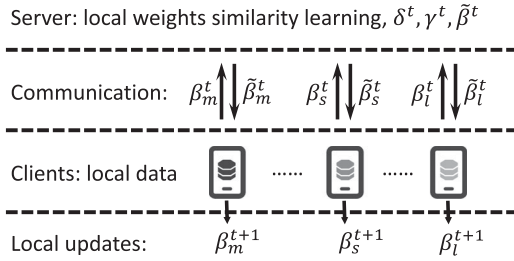


Figure 2. Illustration of PerFL-RSR.

4.2. Proposed Algorithm: PerFL-RSR

In order to design an effective algorithm for the federated client-server system, we address two aspects: (a) communication efficiency; (b) system heterogeneity. To achieve communication efficiency, we design to upload the smallest size of message between clients and the server. For a federated system consists of a large number of clients with variable computational capabilities and network conditions, it is impossible for the server to communicate with all the clients in each communication round. To accommodate system heterogeneity, we design an algorithm that allows low client participation rate in each round.

Communication Efficiency: We design to pass the model weights in the federated system. The update (6) can be decomposed as $\beta^{(t+1)} = \tilde{\beta}^{(t)} - r^{-1} \tau \nabla f_{\sigma}(\beta^{(t)})$, where

$$\tilde{\beta}^{(t)} = r^{-1} \mathbf{H} \beta^{(t)} - r^{-1} \tau \left[-\rho \mathbf{A}^T \delta^{(t)} + \mathbf{A}^T \gamma^{(t)} \right]. \quad (7)$$

Note that, $\delta^{(t)}$ and $\gamma^{(t)}$ are used in the weights similarity learning and hence stored in the server. Therefore, after uploading $\beta^{(t)}$ to the server, $\tilde{\beta}^{(t)}$ can be computed on the server. The gradient $\nabla f_{\sigma}(\beta^{(t)})$ requires local data such that it should be calculated on clients. The model weight for the m th client is located at the $1 + (m-1)p$ to mp coordinates in $\beta^{(t)}$, that is, $\beta_m^{(t)} = (\beta_{1+(m-1)p}^{(t)}, \dots, \beta_{mp}^{(t)})^T$. Then, the model weights could be updated locally as

$$\beta_m^{(t+1)} = \tilde{\beta}_m^{(t)} - r^{-1} \tau \nabla f_{\sigma, m}(\beta_m^{(t)}), \quad (8)$$

where $f_{\sigma, m}(\beta_m^{(t)}) = n_m^{-1} \sum_{i=1}^{n_m} H_{\sigma}(y_{mi} - \mathbf{x}_{mi}^T \beta_m)$.

System Heterogeneity: We enable partial participation in each communication round of the proposed algorithm to adapt system heterogeneity. The participation rate depends on the system condition, such as number of clients, computational capabilities and network conditions. If the number of clients is small, we could allow full participation, and if the number of clients is large and the network condition is poor, we will select part of clients for updating to improve communication efficiency. We address system heterogeneity through random sampling clients for each communication with the server.

Proposed Algorithm: We update the personalized model weight on clients and aggregate all information to investigate the similarity between local weights on the server. The calculations on the server are simple and quick since they are one-step updates and only need matrix multiplication. We summarize our algorithm in Algorithm 1 and illustrate our algorithm in Figure 2. PerFL-RSR selects $|S^{(t)}|$ number of clients for the t th round of communication with the server. The server holds and

updates $\delta^{(t)}$, $\gamma^{(t)}$ and $\tilde{\beta}^{(t)}$. The clients upload local models β_m^t and download information $\tilde{\beta}^{(t)}$ that is required for the next local update. The client uses $\tilde{\beta}^{(t)}$ and the local gradient to calculate $\beta_m^{(t+1)}$. Note that $\beta^{(t+1)} = (\beta_1^{(t+1)T}, \dots, \beta_M^{(t+1)T})^T$. For selected client $m \in S^{(t)}$, the coefficient $\beta_m^{(t)}$ is updated as $\beta_m^{(t+1)}$ and for the not chosen client $m' \notin S^{(t)}$, $\beta_{m'}^{(t+1)} = \beta_{m'}^{(t)}$. Hence, we only specify the update rules of local weights in Algorithm 1. The initial estimator $\beta_m^{(0)}$ is calculated using local data

$$\beta_m^{(0)} = \arg \min_{\beta \in \mathbb{R}^p} f_{\sigma, m}(\beta) + \sum_{j=1}^p p_{\lambda_2}(|\beta_j|). \quad (9)$$

The theoretical requirements of the parameters and the parameter selection for implementation are given in Section 5.

Algorithm 1 Personalized federated learning for robust and sparse regression (PerFL-RSR) method

Require: Tuning parameters $\tau, \rho^0, \alpha, r, \iota, \sigma, \lambda_1, \lambda_2, R$ and number of iteration T ;

- 1: **Initialization:** Set $\gamma^{(0)} = \mathbf{0}$, compute the initial $\beta_m^{(0)}$ by (9) and upload to server
- 2: **for** $t = 0, 1, \dots, T-1$ **do**
- 3: **Server:**
 Update $\delta^{(t+1)}$ by (5), update $\gamma^{(t+1)}$ by (4), and calculate $\tilde{\beta}^{(t)}$ by (7);
 Select $S^{(t)} \subset [M]$;
- 4: **Clients** $m \in S^{(t)}$ **do in parallel:**
 Download $\tilde{\beta}_m^{(t)}$ and calculates $\nabla f_{\sigma, m}(\beta_m^{(t)})$;
 Update $\beta_m^{(t+1)}$ by (8);
 Send $\beta_m^{(t+1)}$ to the server;
- 5: **end for**

Ensure: Client $m \in [M]$ return $\beta_m^{(T)}$ and server return $\beta^{(T)}$.

5. Theoretical Results

In this section, we first show the convergence property of the proposed PerFL-RSR algorithm and then prove the statistical consistency under some mild assumptions.

5.1. Algorithm Convergence

The convergence of the proposed PerFL-RSR algorithm is a non-trivial extension of current ADMM convergence results. Many existing works, such as (Li and Pong 2015; Huang, Chen, and Huang 2019; Lu et al. 2021), considered optimization problems with constraint $\mathbf{A}\eta + \mathbf{B}\beta = \mathbf{0}$ and studied the convergence of the approximated ADMM under assumption either $\mathbf{A}$ is full row rank or the image of matrix $\mathbf{A}$ belongs to the image of $\mathbf{B}$, which are not satisfied for our problem. We will construct the convergence theory for the proposed PerFL-RSR algorithm in another way. First, we show the required assumptions for the loss function and regularization function.

Assumption 1. The gradient of $f(\beta)$ is Lipschitz continuous with the constant $\mu > 0$, such that $\|\nabla f(\beta_1) - \nabla f(\beta_2)\|_2 \leq \mu \|\beta_1 - \beta_2\|_2, \forall \beta_1, \beta_2 \in \mathbb{R}^p$.

Assumption 2. The regularizer $p_\lambda(\cdot)$ is a function such that: (i) It is symmetric around zero and nondecreasing for $t > 0$; (ii) For $t \geq \lambda$, a constant $\iota \in (0, \infty)$, $p_\lambda(t)$ is a constant and $p_\lambda(0) = 0$; (iii) The derivative $p'_\lambda(t)$ exists and is continuous except for a finite number of t with $p'_\lambda(0+) = 1$ and $p'_\lambda(t) \leq \lambda$ for all t ; (iv) For $t > 0$, the function $p_\lambda(t)/t$ is nonincreasing in t ; (v) There exists $\nu > 0$ such that $p_{\lambda,\nu}(t) := p_\lambda(t) + \nu t^2/2$ is convex.

The Huber loss $f_\sigma(\boldsymbol{\beta})$ is Lipschitz continuous with $\mu = \max_{m \in [M], i \in [n_m]} \|\mathbf{x}_{mi}\|_2^2$ and the proof is given in the supplementary material. Conditions (i)–(iii) in **Assumption 2** are generally assumed for regularizers in the literature (Loh and Wainwright 2015; Ma and Huang 2017). Conditions (iv)–(v) in **Assumption 2** are further assumed to derive the cone condition of the estimated weights. The MCP and SCAD regularizers satisfy **Assumption 2** with $\nu = 1/\iota$ and $\nu = 1/(\iota - 1)$, respectively, as discussed in Loh and Wainwright (2015).

To start the convergence analysis, we study the reduction of the augmented Lagrangian after updating $\boldsymbol{\beta}^{(t+1)}$ through Linearized ADMM.

Lemma 1 (Lagrangian Reduction of Updating $\boldsymbol{\beta}$). Let **Assumption 1** hold and $\{\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ be generated by **Algorithm 1**. Let $r > \rho\tau \cdot \varsigma_{\max}(\mathbf{A}^\top \mathbf{A}) + \max(\tau\mu/2, 1)$, then,

$$\begin{aligned} \mathcal{L}_\rho(\boldsymbol{\beta}^{(t+1)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}) - \mathcal{L}_\rho(\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}) \\ \leq - \left[\frac{\varsigma_{\min}(\mathbf{H})}{\tau} + \frac{\rho \cdot \varsigma_{\min}(\mathbf{A}^\top \mathbf{A})}{2} - \frac{\mu}{2} \right] \|\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}\|_2^2. \end{aligned}$$

By the definition of $\mathbf{A}$, we have $\varsigma_{\max}(\mathbf{A}^\top \mathbf{A}) = M + 1$, $\varsigma_{\min}(\mathbf{A}^\top \mathbf{A}) = 1$. The specification of r results in that $\varsigma_{\min}(\mathbf{H})/\tau + \rho \cdot \varsigma_{\min}(\mathbf{A}^\top \mathbf{A})/2 - \mu/2 > 0$.

Since $\mathbf{A}$ is not full row rank, the upper bound of $\|\boldsymbol{\gamma}^{(t+1)} - \boldsymbol{\gamma}^{(t)}\|_2^2$ cannot be given such as in Li and Pong (2015), Huang, Chen, and Huang (2019), and Lu et al. (2021). Instead, we show the convergence through the boundness of parameters $\{\boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ and Lagrangian $\{\mathcal{L}_{\rho^{(t)}}(\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)})\}_{t=1}^\infty$ during the updating process. Then, by the Bolzano-Weierstrass theorem, the sequence $\{\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ must have a convergent subsequence. Therefore, if $\mathcal{L}_\rho(\boldsymbol{\beta}, \boldsymbol{\delta}, \boldsymbol{\gamma})$ is convex, the sequence $\{\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ converges to the unique optimizer. However, in this work, we have a nonconvex optimization problem and we show that there exists a subsequence of $\{\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ that satisfies the KKT conditions for minimization of the augmented Lagrangian.

Theorem 1 (Convergence of Algorithm 1). Let assumptions in **Lemma 1** and **Assumptions 2** hold. If $\rho^{(t+1)} = \alpha\rho^{(t)}$ and $\frac{\min\{\lambda_1^2, \lambda_2^2\}\iota}{2} > \mathcal{L}_0 + \frac{c_\lambda\alpha(\alpha+1)}{2\rho^{(0)}(\alpha-1)}$, where c_λ is a positive constant decided by λ_1 and λ_2 . Assume the selection probabilities for clients are larger than a constant $r_p \in (0, 1)$. Then, $\{\boldsymbol{\beta}^{(t)}, \boldsymbol{\delta}^{(t)}, \boldsymbol{\gamma}^{(t)}\}_{t=1}^\infty$ is bounded and there exists a subsequence $\{\boldsymbol{\beta}^{t_k}, \boldsymbol{\delta}^{t_k}, \boldsymbol{\gamma}^{t_k}\}_{t_k=1}^\infty$, which converges to a stationary point that satisfies the KKT conditions. Further, $\lim_{t \rightarrow \infty} \|\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}\|_2^2 + \|\boldsymbol{\delta}^{(t+1)} - \boldsymbol{\delta}^{(t)}\|_2^2 = 0$, and $\lim_{T \rightarrow \infty} \sum_{t=0}^T \|\boldsymbol{\beta}^{(t+1)} - \boldsymbol{\beta}^{(t)}\|_2^2 \leq \frac{1}{r_p} \left[\frac{\mathcal{L}^0}{c^{(0)}} + \frac{\lambda^2\alpha(\alpha+1)}{2c^{(0)}\rho^0(\alpha-1)} \right]$.

Theorem 1 indicates that the sequence $\{\boldsymbol{\beta}^{(t)}\}_{t=1}^\infty$ converges to a local minimum and that the regression weight $\{\boldsymbol{\beta}^{(t)}\}_{t=1}^\infty$ converges at a linear speed of $\mathcal{O}(\frac{1}{r_p T})$ such that a larger participation rate will fasten the convergence.

5.2. Statistical Consistency

In this section, we study the theoretical properties of the proposed estimator generated by the proposed algorithm PerFL-RSR through analysis of the oracle estimators and local optima of the proposed objective (1).

Theory for the Oracle Estimator To define the oracle estimator, we introduce the following notation. Let $\tilde{\mathbf{X}} = \text{Diag}\{\mathbf{X}_1, \dots, \mathbf{X}_M\} \in \mathbb{R}^{n \times pM}$ be the design matrix corresponding to user-wise regression and $\boldsymbol{\beta}$ be the corresponding model weights. The total sample size is $n = \sum_{m=1}^M n_m$. Next, we define the design matrix when the group structures are known. For each covariate j , the coefficients $(\beta_{1j}, \beta_{2j}, \dots, \beta_{Mj})$ are grouped into K_j groups denoted as $\mathcal{G}_1^j, \dots, \mathcal{G}_{K_j}^j$, where $\mathcal{G}_k^j \cap \mathcal{G}_{k'}^j = \emptyset$ for $k \neq k'$ and $\cup_{k=1}^{K_j} \mathcal{G}_k^j = [M]$. Let ϕ_k^j be the coefficient for group $\mathcal{G}_k^j$, that is for any $m \in \mathcal{G}_k^j$, $\beta_{mj} = \phi_k^j$. Then, there is $K = \sum_{j=1}^p K_j$ total number of parameters. We define $p \times K$ matrix $\mathbf{U}_m$ to link the j th coefficient for client m to its group. Hence, $U_{m,jk'} = 1$ for $m \in \mathcal{G}_{k'}^j$, $k' = \sum_{t=0}^{j-1} K_t + k$ and $U_{m,jk'} = 0$ otherwise. Then, the design matrix given the group structures becomes $\mathbb{X} = \tilde{\mathbf{X}}\mathbf{U} \in \mathbb{R}^{n \times K}$, where $\mathbf{U} = (\mathbf{U}_1^\top, \dots, \mathbf{U}_M^\top)^\top$. The coefficient vector corresponding to $\mathbb{X}$ is $\boldsymbol{\phi} = (\phi_1, \dots, \phi_K)^\top$, where $\phi_{k'} = \phi_k^j$ for $k' = \sum_{t=0}^{j-1} K_t + k$. We simplify the indexing notation of $\mathcal{G}_k^j$ as $\{\mathcal{G}_{k'}\}_{k'=1}^K$ where $\mathcal{G}_{k'} = \mathcal{G}_k^j$ for $k' = \sum_{t=0}^{j-1} K_t + k$. Let $N_{\min}$ represent the minimum sample sizes among all the groups. Let $\boldsymbol{\phi}^*$ be the true model weight and let q be the number of nonzero parameters in $\boldsymbol{\phi}^*$. Without loss of generality, we arrange the nonzero parameters ahead of zeros, and then $\boldsymbol{\phi}^*$ can be expressed as $\boldsymbol{\phi}^{*\top} = (\boldsymbol{\phi}_1^{*\top}, \mathbf{0}^\top)$, where $\boldsymbol{\phi}_1^* \in \mathbb{R}^q$. Accordingly, we arrange $\boldsymbol{\phi}$ and $\mathbb{X}$, such that we write $\boldsymbol{\phi}^\top = (\boldsymbol{\phi}_1, \boldsymbol{\phi}_2)^\top$, where $\boldsymbol{\phi}_1 \in \mathbb{R}^q$, and decompose $\mathbb{X}$ as $\mathbb{X}_1$ and $\mathbb{X}_2$, where $\mathbb{X}_1$ is the first $n \times q$ submatrix of $\mathbb{X}$ and $\mathbb{X}_2$ is the $n \times (K - q)$ submatrix of $\mathbb{X}$.

To conclude the notation, we use $\tilde{\mathbf{X}}$ and $\boldsymbol{\beta}$ to denote the design matrix and model weights for user-wise regression, use $\mathbb{X}$ and $\boldsymbol{\phi}$ to denote the design matrix and model weights with known group structure and use $\mathbb{X}_1$ and $\boldsymbol{\phi}_1$ to denote the design matrix and model weights with both known group structure and sparsity structure.

If the underlying group and sparsity information is known, that is given $\mathbf{U}$ and $\boldsymbol{\phi}_2 = \mathbf{0}$, we define the oracle estimator $\hat{\boldsymbol{\phi}}^\top = (\hat{\boldsymbol{\phi}}_1^\top, \mathbf{0}^\top)$, where

$$\hat{\boldsymbol{\phi}}_1 = \arg \min_{\boldsymbol{\phi}_1 \in \mathbb{R}^q} \sum_{i=1}^n H_\sigma(y_i - \mathbf{z}_i^\top \boldsymbol{\phi}_1), \quad (10)$$

where y_i is the i th element in $(\mathbf{y}_1^\top, \dots, \mathbf{y}_M^\top)^\top$ and $\mathbf{z}_i$ is the i th row of $\mathbb{X}_1$.

Assumption 3. The noise vector $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)^\top$ is a vector of independent random errors with $\mathbb{E}(\epsilon_i | \mathbf{x}_i) = 0$ and $v_{i,\delta} = \mathbb{E}(|\epsilon_i|^{1+\delta}) < \infty$ for all $1 \leq i \leq n$. Let $v_\delta = n^{-1} \sum_{i=1}^n v_{i,\delta} < \infty$, for some $0 < \delta \leq 1$.

Assumption 4. Assume $\varsigma_{\min}(\mathbf{\Sigma}_1) \geq \varsigma_1 N_{\min}/n$, where $\mathbf{\Sigma}_1 = n^{-1} \mathbb{X}_1^\top \mathbb{X}_1$.

Assumption 3 allows heavy-tailed error distributions and does not fix $\mathbb{E}(|\epsilon_i|^{1+\delta})$. Thus, the proposed method works for heterogeneous data. **Assumption 4** is a regular assumption for regression with an additional factor of $n^{-1}N_{\min}$, since we consider personalized models and group clients with the same model weights. Then, the statistical error upper bound of the oracle estimator follows from Sun, Zhou, and Fan (2020).

Proposition 2 (Statistical Accuracy of the Oracle Estimator, Theorem 1 in Sun, Zhou, and Fan (2020)). Let **Assumptions 3** and **4** hold. If $N_{\min} \geq 16\sqrt{2}\varsigma_1^{-1}L_{1z}L_{2z}q^{1/2}t$, where $L_{1z} = \max_{i \in [n]} \|\mathbf{z}_i\|_\infty$ and $L_{2z} = \max_{i \in [n]} \|\mathbf{z}_i\|_2$, and $\sigma = \sigma_0(\nu_\delta n/t)^{1/(1+\delta)}$ for a constant $\sigma_0 \geq 1$, then for any $t > 0$, the oracle estimator $\hat{\boldsymbol{\phi}}_1$ defined in (10) satisfies $\|\hat{\boldsymbol{\phi}}_1 - \boldsymbol{\phi}_1^*\|_2 \leq 4\sigma_0\varsigma_1^{-1}N_{\min}^{-1}nLq^{1/2}\nu_\delta^{1/(1+\delta)}(t/n)^{\delta/(1+\delta)}$, with probability at least $1 - (2q+1)e^{-t}$.

Since we consider personalized regression weights for the clients, our bound differs from the bound in Sun, Zhou, and Fan (2020) by a multiplier $n/N_{\min}$. By letting $t = \log(nq)$, the oracle estimator could achieve an upper bound of order $\mathcal{O}(n(\log n/n)^{\delta/(1+\delta)}/N_{\min})$. Therefore, the estimating error bound depends on the number of data points in each group. If there are similarities in the regression weights between clients, the estimating efficiency is improved by jointly training with all clients. For the square loss and errors with sub-Gaussian tails and finite variance, the upper bound is $\mathcal{O}(\sqrt{n \log n}/N_{\min})$ as given in Ma and Huang (2017), which has the same order as ours for bounded variance case, that is $\delta = 1$.

If only the group information is known, we define the integration oracle estimator as

$$\tilde{\boldsymbol{\phi}} = \arg \min_{\boldsymbol{\phi} \in \mathbb{R}^K} \sum_{i=1}^n H_\sigma(y_i - \mathbf{w}_i^\top \boldsymbol{\phi}) + \sum_{k=1}^K |\mathcal{G}_k| p_{\lambda_2}(|\phi_k|), \quad (11)$$

where $\mathbf{w}_i$ is the i th row of $\mathbb{X}$. The statistical accuracy of $\tilde{\boldsymbol{\phi}}$ from (11) is presented below.

Theorem 3 (Statistical Accuracy of the Integration Oracle Estimator). Let **Assumptions 2–3** hold. Assume $\mathbf{w}_i$'s are sub-Gaussian with $\mathbf{\Sigma}_2 = \mathbb{E}(n^{-1} \mathbb{X}^\top \mathbb{X})$. Let $\nu < \varsigma_{\min}(\mathbf{\Sigma}_2)/3$ and $\varsigma_{\min}(\mathbf{\Sigma}_2) - 3\nu = \varsigma_2 N_{\min}/n$ for constant $\varsigma_2 > 0$. If $\sigma = \sigma_0(\nu_\delta n/\log pM)^{1/(1+\delta)}$ for a constant $\sigma_0 \geq 1$, $c_l \left(\frac{\log pM}{n}\right)^{\delta/(1+\delta)} \leq \lambda_2 \leq \frac{c_u}{R}$, and $n \geq c'q \log pM$, for some constants c_l, c_u and c' , then stationary points $\tilde{\boldsymbol{\phi}}$ of program (11) satisfies the error bound $\|\tilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_2 \leq 6\varsigma_2^{-1}N_{\min}^{-1}nq^{1/2}\lambda_2$, with probability at least $1 - c_1(pM)^{-c_2}$ for some positive constant c_1 and c_2 .

Theorem 3 implies that if we choose $\sigma \asymp \left(\frac{n}{\log pM}\right)^{\frac{1}{1+\delta}}$, $\lambda_2 \asymp \left(\frac{\log pM}{n}\right)^{\delta/(1+\delta)}$, and $n \gtrsim q \log pM$, the integration oracle $\tilde{\boldsymbol{\phi}}$ has an upper bound of $\mathcal{O}(N_{\min}^{-1}nq^{1/2}(\log pM/n)^{\delta/(1+\delta)})$. The details of those constants are given in the proof in the supplementary

material. Next, we show that the oracle estimator $\hat{\boldsymbol{\beta}}^{Or} = U\tilde{\boldsymbol{\phi}}$ is a local minimizer of the proposed objective (1). Let $d_{\min} = \min_{j,k: |\mathcal{G}_k^j| \geq 2} \min_{k_1, k_2 \in \mathcal{G}_k^j, k_1 \neq k_2} |\phi_{k_1}^{*j} - \phi_{k_2}^{*j}|$, where ϕ_k^{*j} is the true common coefficient for the group $\mathcal{G}_k^j$. The $d_{\min}$ is the minimal difference of group-wise coefficients for any covariate.

Theorem 4 (Consistency of the Integration Oracle Estimator). Suppose the assumptions in **Theorem 3** hold. If $[3/2 + 6L_1^2\varsigma_2^{-1}N_{\min}^{-1}nq^{1/2}]\lambda_2 \leq \lambda_1 \leq d_{\min}/t$, where t is given in **Assumption 2**, then, with probability at least $1 - 2(pM)^{-c_0}$, the oracle estimator $\hat{\boldsymbol{\beta}}^{Or}$ is a local minimizer of the proposed objective function (1).

Theory for Local Optima In this part, we analyze the estimator obtained from the proposed PerFL-RSR algorithm, which is a local minimizer of the proposed objective (1). The upper bound and support recovery of local optima for M -estimators with sparsity penalty are studied by Loh and Wainwright (2015) and Loh and Wainwright (2017), respectively. Here, we extend their results to our proposed objective function (1), which includes regularizers for both sparsity and local weights similarity. Let s be the number of nonzero elements in $\boldsymbol{\beta}^*$ and s_G be the number of nonzero elements in $\mathbf{\Omega}\boldsymbol{\beta}^*$, which depends on the group structure.

Theorem 5 (Properties of Local Optima). Let **Assumptions 2–3** hold. Assume $\mathbf{x}_i$'s are sub-Gaussian with covariance matrix $\mathbf{\Sigma}_x$. Let $(3 + 2\varsigma_{\max}(\mathbf{\Omega}^\top \mathbf{\Omega}))\nu < \varsigma_{\min}(\mathbf{\Sigma}_x)$. If $\sigma = \sigma_0(\nu_\delta n/\log pM)^{1/(1+\delta)}$ for a constant $\sigma_0 \geq 1$, $\lambda_1 = r_\lambda \lambda_2$, $c_l \left(\frac{\log pM}{n}\right)^{\delta/(1+\delta)} \leq \lambda_2 \leq \frac{c_u}{R}$, and $n \geq c'q \log pM$, for some constants c_l, c_u and c' , then any stationary points $\hat{\boldsymbol{\beta}}$ of program (1) satisfies the error bound $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \leq \frac{24\lambda_2 \sqrt{(s+s_G)(1+\varsigma_{\max}(\mathbf{\Omega}^\top \mathbf{\Omega}))}}{4a_1 - 3\nu - 2\nu\varsigma_{\max}(\mathbf{\Omega}^\top \mathbf{\Omega})}$, with probability at least $1 - c_1(pM)^{-c_2}$ for some positive constant c_1 and c_2 . Moreover, let $\hat{\mathbf{Q}} = \int_0^1 \nabla^2 f_\sigma(\boldsymbol{\beta}^* + t(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)) dt$ and suppose $\|\hat{\mathbf{Q}}_{SS}^{-1}\|_\infty$ is bounded and $\|\hat{\mathbf{Q}}_{S^c S} \hat{\mathbf{Q}}_{SS}^{-1}\|_\infty \leq \xi < 1$, then with probability at least $1 - c_1(pM)^{-c_2}$, $\text{supp}(\hat{\boldsymbol{\beta}}) \subseteq \text{supp}(\boldsymbol{\beta}^*)$.

Then, given $\sigma \asymp \left(\frac{n}{\log pM}\right)^{\frac{1}{1+\delta}}$, $\lambda_2 \asymp \left(\frac{\log pM}{n}\right)^{\delta/(1+\delta)}$ and $n \gtrsim q \log pM$, we have that for any local minimum $\hat{\boldsymbol{\beta}}$, the upper error bound is $\mathcal{O}(\sqrt{(s+s_G)M}(\log pM/n)^{\delta/(1+\delta)})$.

Tuning Parameter Selection The Huber loss parameter σ and regularization parameters λ_1 and λ_2 have important effects on the proposed PerFL-RSR method. In this part, we specify the tuning method for these three parameters. In the federated system, the client m could choose σ_m using its local data by the variant of Lepski's method (Lepskii 1992) suggested by Sun, Zhou, and Fan (2020). Then, the server will receive local σ_m 's and choose $\sigma = \max_{m \in [M]} \sigma_m$. For the selection of λ_1 and λ_2 , we use the Bayesian information criterion (BIC), which has been widely used for tuning parameter selection (Ma and Huang 2017; Tang and Song 2016). In the federated system, the client m uploaded its local loss $\frac{1}{n_m} \sum_{i=1}^{n_m} H_\sigma(y_{mi} - \mathbf{x}_{mi}^\top \boldsymbol{\beta}_m)$ to the server, and the server calculates the BIC. For given λ_1 and λ_2 , the BIC is defined

as $\text{BIC}(\lambda_1, \lambda_2) = \ln \left\{ \frac{1}{M} \sum_{m=1}^M \frac{1}{n_m} \sum_{i=1}^{n_m} H_\sigma(y_{mi} - \mathbf{x}_{mi}^\top \hat{\boldsymbol{\beta}}_m) \right\} + \frac{\ln n}{n} \cdot \sum_{j=1}^p \hat{K}_j(\lambda_1, \lambda_2)$, where $\hat{K}_j(\lambda_1, \lambda_2)$ is the number of distinct coefficients for variable j corresponding to the estimated $\hat{\boldsymbol{\beta}}$. First, we specify a grid value of λ_2 . As shown in Theorem 5, $\lambda_1 = r_\lambda \lambda_2$, we specify a grid value of r_λ . Then, the parameters λ_1^* and λ_2^* with the smallest BIC value are chosen.

6. Numerical Evaluation

In this section, we evaluate the convergence and statistical theoretical properties of the proposed PerFL-RSR algorithm through synthetic data and real data experiments.

6.1. Synthetic Data Experiments

Data Generation We consider M data units, where each data unit generates data from the linear model: $\mathbf{y}_m = \mathbf{X}_m \boldsymbol{\beta}_m^* + \boldsymbol{\epsilon}_m$, $m = 1, \dots, M$. The design matrix $\mathbf{X}_m$ is generated from the multivariate normal distribution with mean 0 and $\text{cov}(x_{is}, x_{it}) = 0.3$, $s \neq t$, $\text{cov}(x_{it}, x_{it}) = 1$, $s, t \in [p]$. We divide the data units into two groups: group one consists of units $\{1, \dots, M/2\}$; group two consists of units $\{M/2 + 1, \dots, M\}$. The model weight is $(1, 3, \mathbf{0}_{p-2}^\top)^\top$ for group one and $(2, 3, \mathbf{0}_{p-2}^\top)^\top$ for group two. We consider The number of observations is equal for all units, that is $n_1 = \dots = n_M$.

We specify the choices of tuning parameters: $\tau, \rho^{(0)}$ and r for the linearized ADMM, parameter ι for SCAD and MCP as follows. Following a similar strategy for ADMM algorithms (Ma and Huang 2017; Lu et al. 2021), we fix τ as 1 and $\rho^{(0)}$ as 2. Let the rank parameter be $r = \rho^{(0)} \tau \cdot \varsigma_{\max}(\mathbf{A}^\top \mathbf{A}) + \max\{\tau \mu/2, 1\}$. For the regularization function, let $\iota = 3$.

We will conduct four sets of experiments to evaluate the proposed algorithm: (a) the effect of grouping to check that PerFL-RSR is able to recover the structure of model weights; (b) the effect of heavy-tailed noise for the robustness of the proposed PerFL-RSR algorithm; (c) the effect of tuning parameters for convergence of the proposed PerFL-RSR algorithm, including Huber blending parameter σ and regularization parameter λ_1 and λ_2 ; (d) the effect of partial participation of clients in each communication round.

6.1.1. Effect of Grouping

In this set of experiments, we compare the efficiency in terms of estimation accuracy and coefficient structure recovery of different estimators.

Compared Methods: (a) The proposed method (PerFL-RSR): using fusion and sparsity penalized Huber loss on the federation of data units, which is able to recover the true structure of the model weights; (b) The client-specific method (Individual): using sparsity penalized Huber loss on each unit, which is able to recover the sparsity of the model weights but can not detect the group structure of the weights; (c) FedAvg (McMahan et al. 2017): using sparsity penalized Huber loss on the federation of data units, which misspecifies the model.

Performance Measurements: In the experiment with the synthetic dataset, we evaluate the performance of the three methods in terms of the following measurements: (a) MSE: mean square

error of the estimated weights; (b) Recall: the fraction of correctly identified nonzero weights over true sparsity; (c) Precision: the fraction of correctly identified nonzero weights over total identified nonzero one; (d) F_1 -score: the sparsity recovery that is defined as $F_1\text{-score} = 2(\text{recall}^{-1} + \text{precision}^{-1})^{-1}$, where the F_1 score ranges between 0 and 1, and the closer to 1, the better the sparsity recovery; (e) AMS: average model size that is defined as the number of distinct values in the estimated weights.

In this simulation, the error terms are from the Student's t distribution: $t(1.5)$ for group one and $t(3)$ for group two. We fix the dimension at $p = 15$, the Huber blending parameter, and the regularization parameters. The simulation is repeated 50 times. We specify a grid value of λ_2 . According to Theorem 5 that $\lambda_1 = r_\lambda \lambda_2$, we specify a grid value of $r_\lambda = \{3, 6, 9\}$. The parameters λ_1^* and λ_2^* with the smallest BIC value are chosen. Table 1 reports the results for the three methods with MCP and SCAD regularizers. We can observe that the proposed method has the best model performance in terms of estimation accuracy (MSE), sparsity recovery (TP to F_1 -score), and group structure detecting (AMS). The FedAvg method calculated one global model for all clients, which misspecified the true model, and hence, FerAvg has the largest MSEs. The Individual method calculated the local model for each client so that it could recover sparsity but it can not shrink the AMS. For the case with $M = 10$ and $n_m = 50$, the Individual method uses 50 data points for estimating a coefficient, while the PerFL-RSR method uses at least 250 data points for estimating a coefficient. Therefore, the proposed method takes advantage of jointly training data and significantly reduces the MSE compared with the Individual method. Moreover, the proposed method successfully detected the similarity between local weights and resulted in an AMS close to the true model size 4.

6.1.2. Effect of Heavy-Tailed Noise

The proposed PerFL-RSR algorithm is designed for data possibly contaminated with heavy-tailed noises or extreme values, which are pervasive in the real world. In this set of experiments, we aim to compare the practical performance of the proposed Huber-based loss with ℓ_2 based loss under different choices of noise distributions. For the training with ℓ_2 loss, the proposed algorithm is used, where the gradient of the Huber loss is replaced by the gradient of ℓ_2 loss. Hence, we denote the personalized federated learning with ℓ_2 loss as PerFL-l2. We consider the following three noise distributions: (a) Normal: for group one $e_i \sim N(0, 1)$ and for group two $e_i \sim N(0, 0.5^2)$, where the distributions have light tails and finite moments; (b) Cauchy: for group one $e_i \sim \text{Cauchy}(1)$ and for group two $e_i \sim \text{Cauchy}(1.5)$, where the distributions have heavy tails and infinite variance; (c) Student's t : for group one $e_i \sim t(1.5)$ and for group two $e_i \sim t(3)$, where the distributions have tails between Normal and Cauchy distributions.

Figure 3 presents the results of the logarithm of the MSEs under three contexts where the shadow areas correspond to the interquartile ranges (IQR) that include the middle 50% of simulated results. The first row in Figure 3 presents results of the logarithm of the MSEs versus different dimensions p with fixed unit sample size $n_m = 100$, and the total number of units

Table 1. The results of comparison between different methods based on 50 replications.

Method	Regularizer	(M, n_m)	MSE	Recall	Precision	F_1 score	AMS
Individual	MCP	(10,50)	3.21(0.27)	0.99	0.93	0.95	24.78(0.65)
Individual	SCAD	(10,50)	3.12(0.29)	0.99	0.93	0.95	24.32(0.55)
FedAvg	MCP	(10,50)	16.15(0.04)	0.64	0.51	0.54	3.84(0.17)
FedAvg	SCAD	(10,50)	16.01(0.03)	0.61	0.54	0.55	3.54(0.15)
PerFL-RSR	MCP	(10,50)	1.10(0.20)	1.00	0.94	0.96	4.28(0.10)
PerFL-RSR	SCAD	(10,50)	1.09(0.20)	1.00	0.96	0.97	4.22(0.09)
Individual	MCP	(10,100)	2.29(0.25)	1.00	0.91	0.93	27.34(0.89)
Individual	SCAD	(10,100)	2.46(0.28)	1.00	0.90	0.92	28.08(0.93)
FedAvg	MCP	(10,100)	15.89(0.01)	0.54	0.83	0.63	2.46(0.09)
FedAvg	SCAD	(10,100)	15.86(0.01)	0.54	0.87	0.65	2.38(0.09)
PerFL-RSR	MCP	(10,100)	0.76(0.20)	1.00	0.97	0.98	4.14(0.06)
PerFL-RSR	SCAD	(10,100)	0.69(0.16)	1.00	0.99	0.99	4.10(0.05)
Individual	MCP	(50,100)	6.79(0.44)	1.00	0.91	0.93	135.02(2.20)
Individual	SCAD	(50,100)	7.06(0.48)	1.00	0.90	0.92	138.08(2.24)
FedAvg	MCP	(50,100)	35.38(0.00)	0.51	1.00	0.67	2.02(0.02)
FedAvg	SCAD	(50,100)	35.37(0.00)	0.50	1.00	0.67	2.00(0.00)
PerFL-RSR	MCP	(50,100)	3.26(0.69)	1.00	0.97	0.97	4.46(0.18)
PerFL-RSR	SCAD	(50,100)	3.25(0.69)	1.00	0.95	0.96	4.50(0.20)

NOTE: The numbers in the parentheses are the standard deviations based on 50 replications (value 0.00 means less than 0.005).

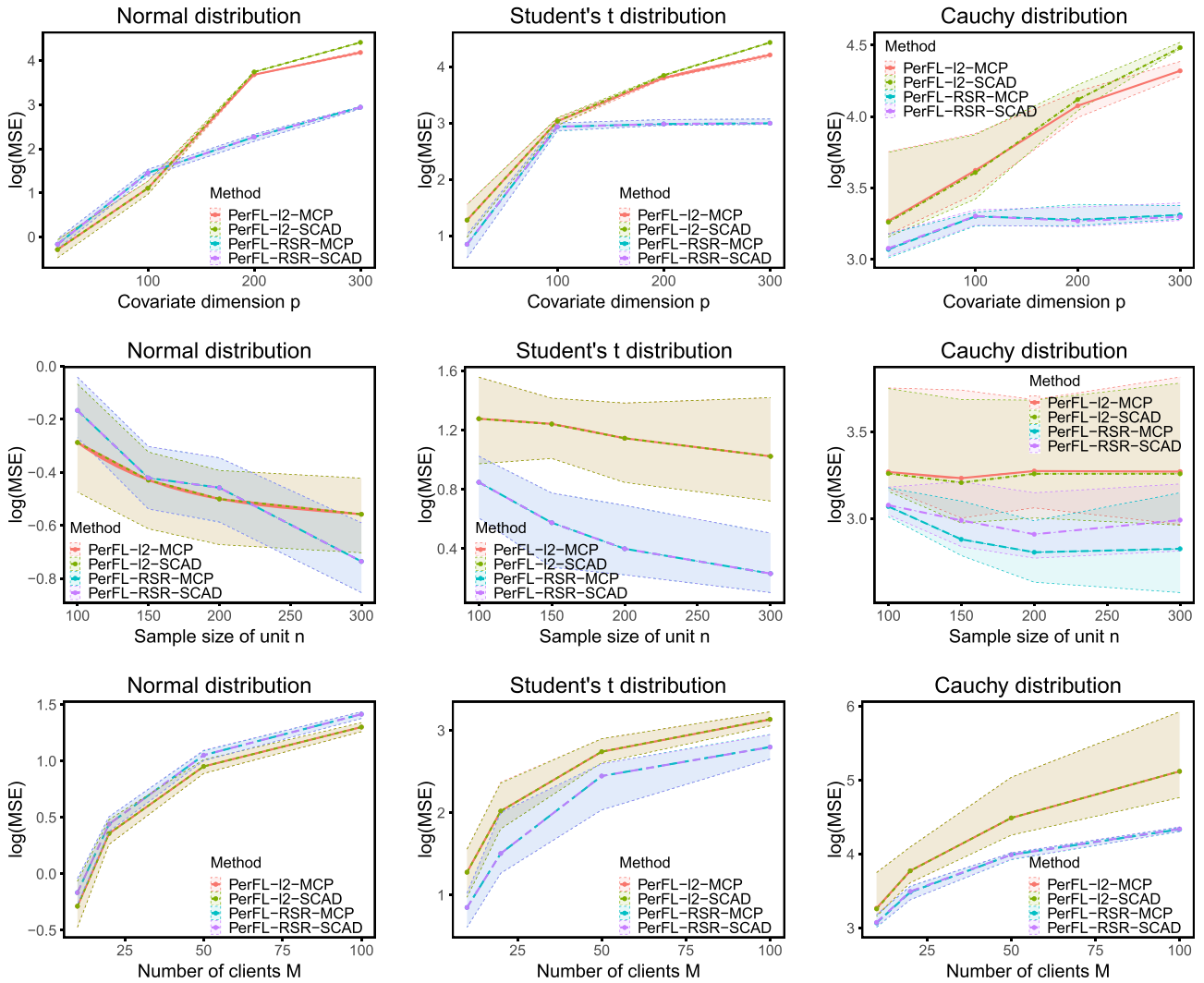


Figure 3. Simulation results over different dimensions p , unit sample sizes n_m and total number of units M for three noise distributions. The shadow areas correspond to the interquartile ranges.

$M = 10$ for three noise distributions. The second row in Figure 3 presents the results of the logarithm of the MSEs versus different unit sample size n_m with fixed dimensions $p = 15$ and the total number of units $M = 10$ for three noise distributions. The

third row of Figure 3 presents the results of the logarithm of the MSEs versus different unit sample size n_m with fixed dimensions $p = 15$ and the total number of units $M = 10$ for three noise distributions.

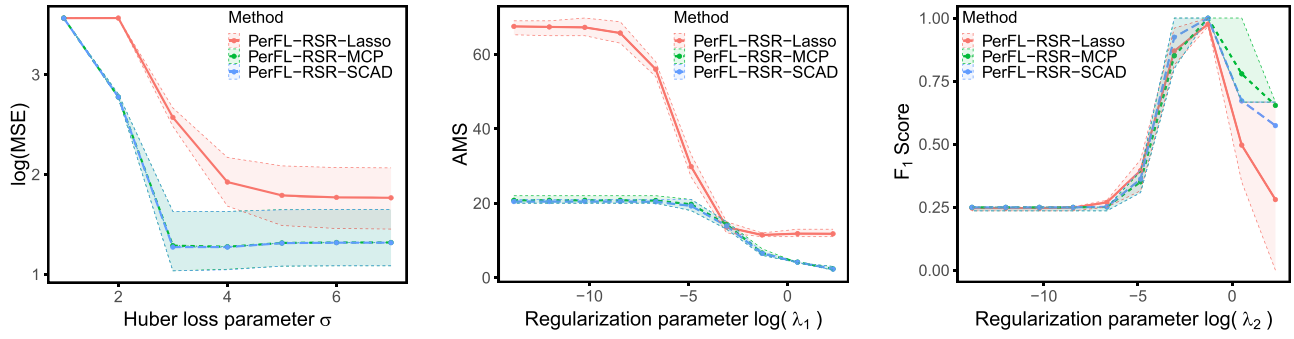


Figure 4. Simulation results over different Huber loss blending parameter σ , regularization parameters λ_1 , and λ_2 . The shadow areas correspond to the interquartile ranges.

For the light-tailed, that is Normal distribution case, the Huber loss performs better than the square loss when increasing the dimension p as we can observe that the MSEs of the square loss increase faster than that of the Huber loss. The Huber loss and the square loss have a similar decreasing speed of MSE as the sample size increases, which should be $\mathcal{O}(n^{-1/2})$. The Huber loss and the square loss also have a similar increasing speed of MSE as the number of clients increases, which should be $\mathcal{O}(M^{1/2})$. For the Student's t distribution, where the tail is considered heavy, the Huber loss performs better than the square loss in all cases. For the Cauchy distribution, where the tail is considered to be heavier than the Student's t distribution, we can observe larger differences between the MSEs from the Huber loss and ℓ_2 loss, which is also consistent with the theory because heavy-tailed noise breaks the finite second-moment assumption for the square loss method. The patterns of MSE changes as dimension p , the unit sample size n , and the number of units M increase are consistent with our theory given in Theorem 5. We can observe that the proposed method using Huber loss outperforms the method using square loss in terms of both MSE and IQR of the MSEs.

6.1.3. Effect of Tuning Parameters

Figure 4 presents results for different Huber loss blending parameters σ , regularization parameters λ_1 , and λ_2 . We can observe that it is important to choose the tuning parameters for better fitting performance. First, small σ implies more similarity with ℓ_1 loss, which may suffer from slow convergence, and large σ leads to more similarity to square loss, which is not suitable for heavy-tailed error. Besides, the group structure of the weights is sensitive to the fusion parameter λ_1 . Figure 4 shows that as λ_1 increases, AMS decreases. Large λ_1 leads to more equal values in the estimates and a decrease of the AMS while small λ_1 leads to more personalization instead of generalization. As shown in Figure 4, the regularization parameter λ_2 controls the sparsity recovery. Small λ_2 shrinks fewer weights to zero, and large λ_2 forces more weights to zero.

6.1.4. Effect of Participation Proportion

The proposed PerFL-RSR addresses system heterogeneity. That means that we allow partial participation in each round of communication between the server and clients.

In this set of experiments, we study the effect of the proportion of clients' participation during each round of communication with the server. Figure 5 shows the convergence results of

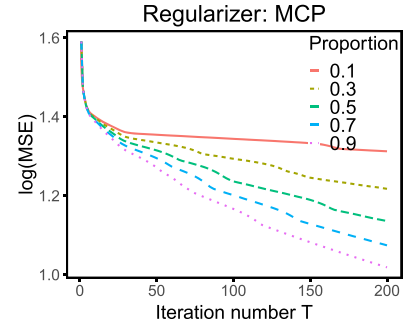


Figure 5. Simulation results over participation proportions.

the proposed estimator with the MCP regularizer. The results for the SCAD regularizer are similar. The plot to the left of Figure 5 shows the logarithm of MSE versus the number of iterations T . It shows that with a larger participation proportion, the estimated weights converge faster in terms of the number of iterations. Because there are more clients participating in the information update.

6.2. Real Data Analysis

Dataset Description In this section, we study the performance of the proposed PerFL-RSR algorithm on the Communities, and Crime dataset from the UCI Machine Learning Repository Redmond (2011). We believe that there is a linear relationship between demographic variables (99 features) and the total number of violent crimes per 100K population (ViolentCrimesPerPop). Therefore, our objective is to estimate the linear model as well as identify the demographic variables with significant effects. We randomly split the data into training data which includes 80% of data points, and test data which includes the remaining 20% of data points. We repeat the split and prediction process 10 times.

The Communities and Crime dataset collected data from communities across the United States. The county information has about 55% missing in the original dataset, and we recover all the county information through the community names. We treat each community as an independent client. We select counties with more than three data points. Then, we have 111 clients in total. The number of observations in each county ranges from 4 to 76, and we summarize the number of observations in each state in Table 2. For counties with few observations, the local regression models will suffer from poor model performance due

Table 2. Summary of the number of observations in each county (Min, Q1, Q3, and Max. stand for minimum, first quantile, third quantile, and maximum, respectively).

	Min.	Q1	Median	Mean	Q3	Max.
n_m	4	5	7	9.5	12	76

Table 3. Results of different methods for the Communities and Crime dataset.

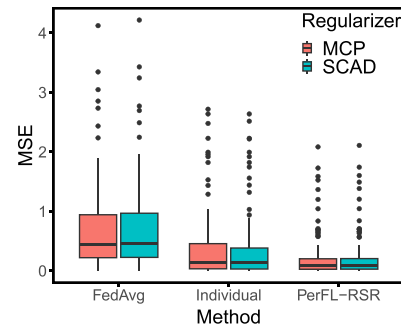
Method	Regularizer	MSE	SE	Size
Individual	MCP	0.54(0.03)	1.75(0.55)	847.80(11.85)
	SCAD	0.54(0.03)	1.72(0.54)	858.00(12.72)
FedAvg	MCP	0.83(0.03)	1.50(0.47)	81.70(1.07)
	SCAD	0.84(0.03)	1.50(0.48)	79.60(1.11)
PerFL-RSR	MCP	0.39(0.02)	1.30(0.41)	67.70(0.54)
	SCAD	0.39(0.02)	1.30(0.41)	67.20(0.36)

NOTE: The numbers in the parentheses are the standard deviations based on 10 replications.

to a lack of data. Therefore, the proposed PerFL-RSR estimator should be employed to improve the total model performance and enable personalized models.

We compare three algorithms: the client-specified method (Individual), FedAvg, and the proposed PerFL-RSR algorithm. To address robustness, we use Huber loss for all methods. Both the MCP and the SCAD regularizers are used. We evaluate the method's performance by the following measurements: (a) MSE: the mean of the test mean square errors across all 111 counties; (b) SE: the standard error of the test mean square errors across all 111 counties; (c) Size: the sum of numbers of nonzero coefficients for each coordinate. Table 3 shows the performance of all methods based on 10 replications. PerFL-RSR estimators achieve the best model performance in terms of MSE and SE. More specifically, the PerFL-RSR estimators reduce the MSE by at least 27.78% as well as reduce the model size by at least 92.01% compared with the individual models, and the PerFL-RSR estimators reduce the MSE by at least 53.01% as well as reduce the model size by at least 17.58% compared with FedAvg. Moreover, the PerFL-RSR estimators reduce the standard errors of MSEs by at least 24.42% compared with the individual models. FedAvg has similar SEs with the PerFL-RSR since it uses all the data to jointly train a global model. Thus, using the PFL framework, we make full use of the data from all users, which enables us to improve personalized model performance. Note that for 111 counties and 100 demographic features, the total number of coefficients to be estimated is 11,100. The individual model with either MCP or SCAD regularizer could select features with significant effects and results in 7.5 significant weights on average for each county. However, FedAvg aggregated weights from all counties and failed to select nonzero weights since the nonzero weights are in different coordinates for different counties. The PerFL-RSR estimators with MCP and SCAD regularizers contain only 67.70 and 67.20 different nonzero weights, which means the PerFL-RSR estimator shrinks the total model size significantly.

We also examine the MSEs for each county, and the boxplots of MSEs across counties for different methods are presented in Figure 6. FedAvg has the worst performance with the largest MSE on average and the most spread MSEs, which means that the model performance for each client is worse than the other two methods. The Individual method is better than FedAvg by

**Figure 6.** Boxplots of MSEs across counties for different methods.

using personalized models. Some counties still have poor individual model performance because of a lack of observations. The PerFL-RSR method further improves the model performance of the Individual method through joint training with data from all counties. Thus, it shows that the large MSEs in the Individual method are reduced.

7. Conclusion

In this work, we study personalized federated learning for robust regression with sparsity. We propose the regression weights obtained by minimizing a Huber loss with the fused sparse penalty. Next, we develop an approximated ADMM algorithm in the federated learning framework. Therefore, our proposed method takes into consideration both personalization and generalization and addresses both statistical heterogeneity and system heterogeneity. We show that the proposed PerFL-RSR reaches a convergence rate of $\mathcal{O}(1/T)$ and prove upper error bound of any local optima obtained from the proposed PerFL-RSR method.

Supplement Materials

Supplement.pdf: Additional theoretical results and detailed proofs for all theorems and corollaries in this article (pdf file).

PerFL-RSR-experiments: The codes and data to implement and reproduce the simulation and real data results.

Disclosure Statement

The authors report there are no competing interests to declare.

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